DBMS Notes

Sample Study Material for RAG Testing

Introduction to DBMS

A Database Management System, or DBMS, is software used to create, store, organize, retrieve, and manage data. Examples of database systems include relational databases.

Database and Tables

A database contains related data. In a relational database, data is commonly organized into tables. A table contains rows, also called records or tuples, and columns, also called attributes.

Primary Key

A primary key uniquely identifies each record in a table. A primary key should contain unique values and should not be null.

Foreign Key

A foreign key is a field that refers to a key in another table. It helps establish relationships between tables.

Normalization

Normalization is the process of organizing data to reduce redundancy and improve data integrity. It usually divides data into related tables and establishes relationships between them.

First Normal Form

First Normal Form, or 1NF, requires atomic values and eliminates repeating groups. Each field should contain a single value.

Second Normal Form

Second Normal Form, or 2NF, requires the table to be in 1NF and removes partial dependency, meaning non-key attributes should depend on the whole primary key.

Third Normal Form

Third Normal Form, or 3NF, requires the table to be in 2NF and removes transitive dependency, so non-key attributes should depend directly on the primary key.

SQL

Structured Query Language, or SQL, is used to create, retrieve, update, and delete data. Common commands include SELECT, INSERT, UPDATE, and DELETE.

Example

A Student table may contain StudentID, Name, and Department. StudentID can be the primary key.